

Data Mining

Lecture 4 @FII

- Outlier detection

Noise and Outliers in Data

Noise refers to **random errors or variability in data** that do not represent the true underlying phenomenon.

- Typical causes:
 - measurement errors
 - sensor inaccuracies
 - data entry mistakes
 - transmission errors
- Noise **usually** appears as **small random fluctuations around normal values**.
- We will insist more on noise reduction in a future lecture

Outliers are **data objects that deviate significantly from the majority of the data**

- They may indicate:
 - rare events
 - abnormal behavior
 - system faults
 - fraud or anomaly
- Outliers may contain **important information** and therefore are often **analyzed rather than removed**.

Noise and Outliers in Data

Noise

- Represents random error in data
- Usually caused by measurement errors
- Typically removed during preprocessing

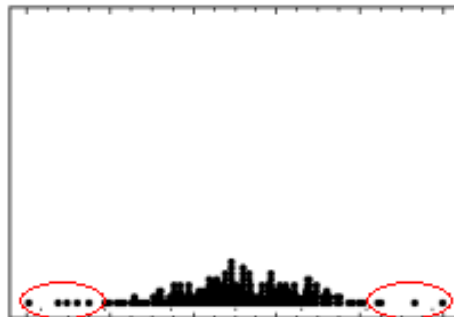
Outlier

- Extreme or unusual observation
- May represent a real but rare event
- Often detected and analyzed, but may also be removed in a data cleaning process

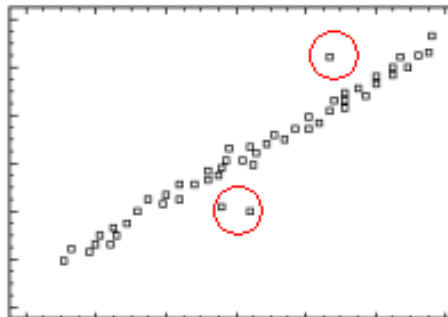
- Both **noise and outliers may appear as unusual data values.**
- All noise may appear as outliers, but not all outliers are noise.
- Many outlier detection algorithms identify **both anomalies and potential noise.**

Outlier definition

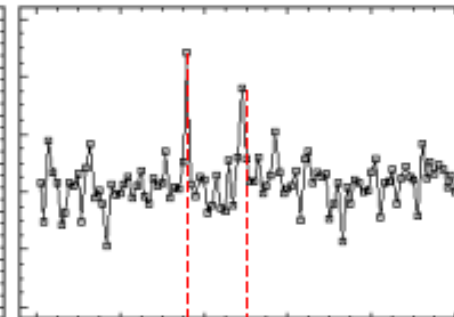
- *An observation that deviates so much from other observations as to arouse suspicions that it was generated by a different mechanism. [Hawkins, 1980]*
- *An outlier is an observation or subset of observations which appear to be inconsistent with the remainder of that dataset [Barnet & Lewis, 1994]*
- *An outlier is an observation that lies outside the overall pattern of a distribution [Moore & McCabe, 1999]*
- *An outlier in a set of data is an observation or a point that is considerably dissimilar or inconsistent with the remainder of the data [Ramaswamy, Rastogi & Shim, 2000]*
- *Outliers are those data records that do not follow any pattern in an application. [Chen, Tan & Fu, 2003]*



(a) Outliers w.r.t. a distribution



(b) Outliers w.r.t. a pattern



(c) time series outliers

Outlier detection

Outlier detection (also called anomaly detection) is the process of identifying data objects that deviate significantly from the majority of the data.

These objects are called:

- *Outliers*
- *Anomalies*
- *Exceptions*
- *Deviations*

Why are outliers important?

- They may indicate errors in data collection
- They may represent rare or abnormal events
- They can reveal interesting patterns in the data

Applications of Outlier Detection

Fraud Detection

- Unusual credit card transactions
- Insurance fraud

Cybersecurity

- Intrusion detection in networks
- Suspicious login behavior

Industrial Monitoring

- Fault detection in machines
- Sensor anomalies

Healthcare

- Abnormal medical measurements
- Disease detection

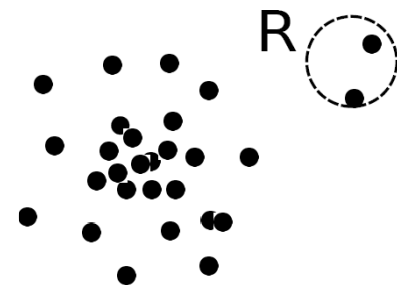
Data Cleaning

- Identifying errors in datasets

Types of Outliers

Global outlier (or point anomaly)

- A single data object that is far from the rest of the data.
- Example: One transaction of \$10,000 among transactions of \$10–\$200.

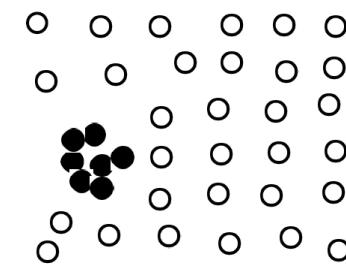


Contextual outlier (or conditional outlier)

- An observation that is abnormal in a specific context.
- Attributes of data objects should be divided into two groups
 - Contextual attributes: defines the context, e.g., time & location
 - Behavioral attributes: characteristics of the object, used in outlier evaluation, e.g., temperature
- Can be viewed as a generalization of *local outliers*—whose density significantly deviates from its local area
- Example: Temperature of 25°C in winter may be normal, but 25°C in a polar region may be abnormal.

Collective Outliers

- A group of data objects that are abnormal together, even if individual points may appear normal.
- Example: A sequence of unusual network traffic patterns.



Challenges in Outlier Detection

- Modeling normal objects and outliers properly
 - There is often no clear definition of an outlier
 - Hard to enumerate all possible normal behaviors in an application
 - The border between normal and outlier objects is often a gray area
 - Data may be high-dimensional
- Application-specific outlier detection
 - Choice of distance measure among objects and the model of relationship among objects are often application-dependent
 - E.g., clinic data: a small deviation could be an outlier; while in marketing analysis, larger fluctuations
- Handling noise in outlier detection
 - Noise may distort the normal objects and blur the distinction between normal objects and outliers. It may help hide outliers and reduce the effectiveness of outlier detection
- Understandability
 - Understand why these are outliers: justification of the detection
 - Specify the degree of an outlier: the unlikelihood of the object being generated by a normal mechanism

Outlier Detection - methods

Statistical Methods

- Assume an underlying statistical distribution of the data and identify points that deviate significantly from it.
 - Z-score, Interquartile Range (IQR), Mahalanobis Distance, Stahel–Donoho Outlyingness and Extensions

Distance-Based Methods

- Measure how far a data point is from its neighbors. Points with large distances are considered outliers.
 - k-Nearest Neighbor (kNN) distance, Distance to k-th nearest neighbor, (ϵ, π) -model

Density-Based Methods

- Compare the local density of a point with the density of its neighbors. Points in low-density regions are potential outliers.
 - Local Outlier Factor (LOF), Local Density Deviation, Connectivity-based Outlier Factor

Machine Learning Methods

- Learn patterns of normal data and detect instances that do not fit the learned model.
 - DBSCAN (noise points), Small-cluster based detection, Cluster distance analysis
 - Isolation Forest, One-Class SVM, Autoencoders (Deep Learning based anomaly detection)

Outlier Detection - Labeling versus scoring

With respect to the output of an outlier detection algorithm

- Labeling approaches
 - Binary output
 - Data objects are labeled either as normal or outlier
- Scoring approaches
 - Continuous output
 - For each object an outlier score is computed (e.g. the probability for being an outlier)
 - Data objects can be sorted according to their scores
- Many scoring approaches focus on determining the top- n outliers (parameter n is usually given by the user)
- Scoring approaches can usually also produce binary output if necessary (e.g. by defining a suitable threshold on the scoring values)



Statistical methods for outlier detection

Statistical outlier detection

Statistics-based intuition

- Normal data objects follow a “generating mechanism”, e.g. some given statistical process
- Abnormal objects deviate from this generating mechanism
- Most methods assume some statistical distribution and find records which deviate significantly from the assumed model.

Univariate outliers – z score

Idea

- The Z-score method measures **how far a data point is from the mean of the dataset**, expressed in terms of standard deviations.

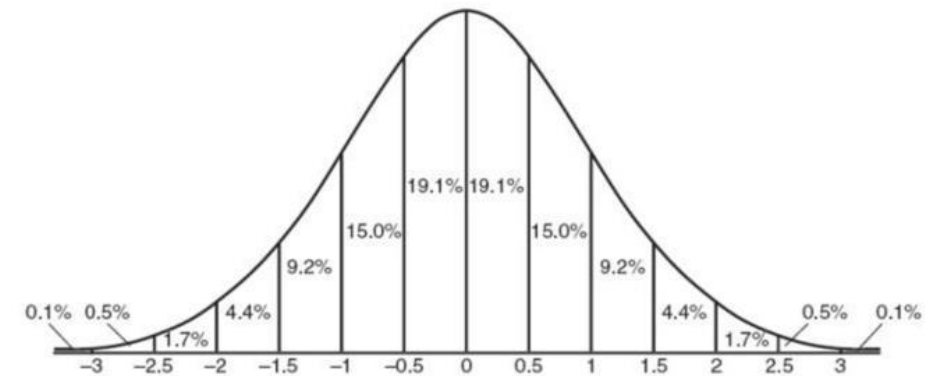
Assumption: normal (Gaussian) distribution

Z-score

$$z = \frac{x - \bar{x}}{s}$$

Common ‘thumb-rule’ thresholds : an observation is considered an outlier if it is more than 2.5 (**1.24%**), 3 (**0.27%**), 3.5 (**0.0466%**) standard deviations far from the mean of the distribution

Ex: $|z| > 3 \rightarrow$ possible outlier



Note: $\bar{x}$ and s are themselves very sensitive to outliers. Extreme values skew the mean

Robust z score

Problem with z-score

- Extreme values can **shift the mean and inflate the standard deviation**

As a result:

- the Z-score of an outlier may become **smaller than expected**
- some outliers may **not be detected**

Solution:

- mean $\rightarrow$ median: $\tilde{x} = \text{median}(x_1, x_2, \dots, x_n)$
- standard deviation $\rightarrow$ MAD (Median Absolute Deviation):

$$\text{MAD} = b \times \text{median}(|x_i - \tilde{x}|)$$

where $b = 1.4826$ if the distribution is Gaussian

Modified Z-Score

$$z_i^* = (x_i - \tilde{x}) / \text{MAD}$$

The constant $b = 1.4826$ makes the statistic comparable to the standard Z-score under normality.

Decision rule:

$|z_i^*| > 3.5 \rightarrow$ potential outlier

What if data is not Gaussian?

Simple idea:

Transform the variable to induce normality

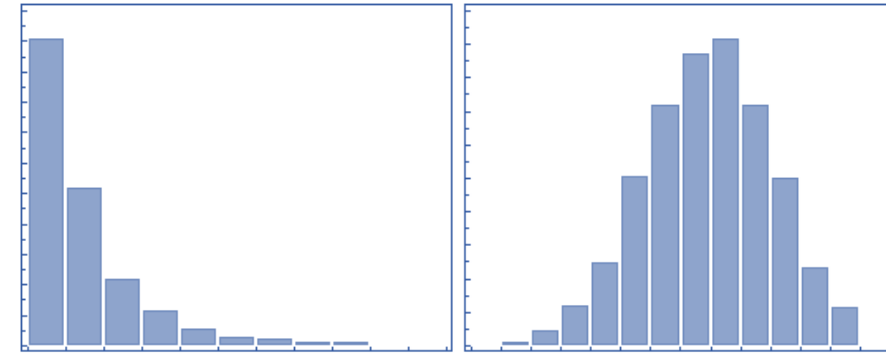
The easiest transformation relies on taking the logarithm of the variable of interest (a special case of the *Box-Cox* transformation)

- The log “squeezes” large values more

For each observation (x), compute:

$$y = \log(x)$$

After the transformation, the distribution often becomes **more symmetric and closer to normal**.

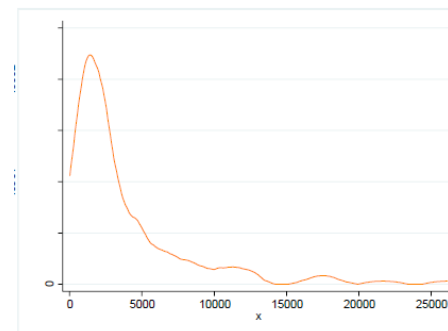


When to Use Log Transformation

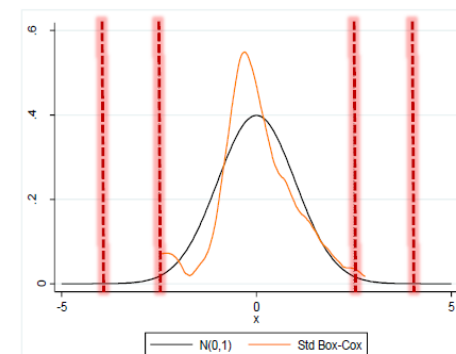
- strongly **right-skewed distributions**
- **positive-valued data**
- variables spanning **multiple scales**

Typical examples: income, transaction amounts, city sizes, website traffic.

Raw untransformed data



Transformed data



Univariate outliers – IQR rule (Tukey)

Idea

- The Interquartile Range (IQR) method identifies outliers based on the spread of the **middle 50% of the data**.
- It is a **robust statistical method** because it is not strongly affected by extreme values.

1. Compute Quartiles

Q_1 – first quartile (25th percentile)

Q_2 – median (50th percentile)

Q_3 – third quartile (75th percentile)

2. Compute the Interquartile Range - measures the spread of the central portion of the data

$$IQR = Q_3 - Q_1$$

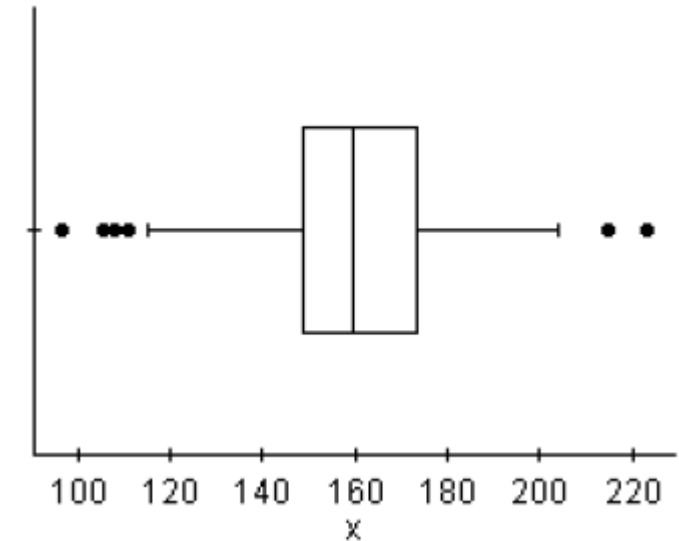
Decision Rule

A data point x is considered an outlier if:

$$x < Q_1 - 1.5 \cdot IQR$$

or

$$x > Q_3 + 1.5 \cdot IQR$$



Univariate outliers – IQR rule (Tukey)

Why $1.5 \cdot \text{IQR}$?

For a normal distribution:

$$Q_1 \approx -0.674\sigma$$

$$Q_3 \approx 0.674\sigma$$

$$\text{IQR} \approx 1.349\sigma$$

Upper bound:

$$Q_3 + 1.5 \cdot \text{IQR} \approx 0.674\sigma + 1.5 \cdot 1.349\sigma \approx \mathbf{2.7\sigma}$$

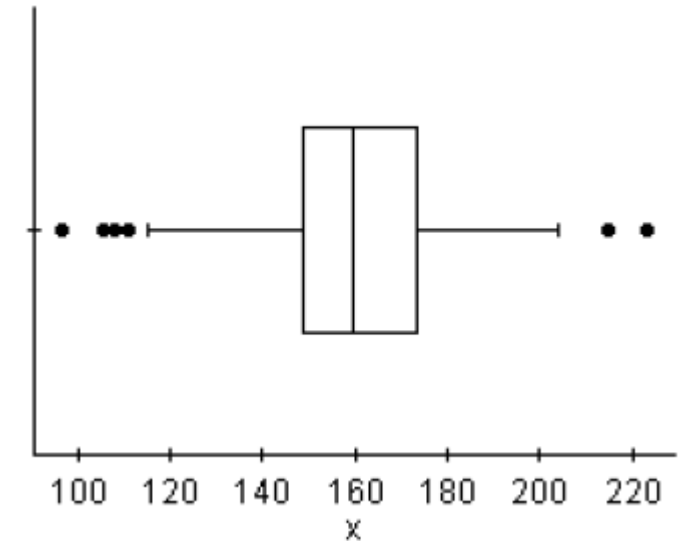
$$\text{Lower bound} \approx \mathbf{-2.7\sigma}$$

Thus, the IQR rule roughly corresponds to $\mu \pm 2.7\sigma$

In a normal distribution:

- about **99.3% of observations** lie within $\pm 2.7\sigma$
- only about **0.7% of values** fall outside this range

For extreme outliers use $Q_1 - 3 \cdot \text{IQR}$, $Q_3 + 3 \cdot \text{IQR}$ ($\pm 4.7\sigma \rightarrow 3 \times 10^{-6} \%$)



The IQR rule remains **robust and distribution-free**, which is why it is widely used in **boxplots** and **exploratory data analysis**.

Statistical Hypothesis Tests (Grubbs Test)

Idea

- The **Grubbs test** is a statistical hypothesis test used to detect **one outlier in a univariate dataset** that is approximately **normally distributed**.
- It evaluates whether the most extreme value in the sample is significantly different from the others.

Hypotheses

H_0 : The sample contains **no outliers**

H_1 : The sample contains **one outlier**

Test Statistic - the **largest standardized deviation from the sample mean**

$$G = \max |x_i - \bar{x}| / s$$

Decision Rule

Extract G_{crit} from Grubbs tables (depends on sample size **n** and significance level **α**).

If $G > G_{\text{crit}} \rightarrow$ reject $H_0 \rightarrow$ the value is an **outlier**

Note

- Detects **one outlier at a time**. After removing a detected outlier, the test may be applied again

Statistical Hypothesis Tests (Grubbs Test)

$$G_{crit} = \frac{(n-1)}{\sqrt{n}} \sqrt{\frac{t_{\alpha/(2n), n-2}^2}{n-2 + t_{\alpha/(2n), n-2}^2}}$$

For $\alpha = 0.05$:

$n=5 \rightarrow G_{crit} \approx 1.67$

$n=10 \rightarrow G_{crit} \approx 2.29$

$n=50 \rightarrow G_{crit} \approx 3.13$

$n=100 \rightarrow G_{crit} \approx 3.32$

$n=1000 \rightarrow G_{crit} \approx 4.02$

Grubbs $\approx$ Z-score with a threshold adapted to the sample size.

Multivariate Outliers: Mahalanobis Distance

When data contain **multiple variables**, outliers cannot be detected reliably using univariate methods such as the Z-score.

A common approach is to use the **Mahalanobis distance**, which measures how far a point is from the **center of a multivariate distribution**, taking into account the **correlations between variables**.



Multivariate Outliers: Mahalanobis Distance

- For an observation vector $\mathbf{x}$, the squared Mahalanobis distance is defined as:

$$D^2 = (\mathbf{x} - \bar{\mathbf{x}})^T \mathbf{S}^{-1} (\mathbf{x} - \bar{\mathbf{x}})$$

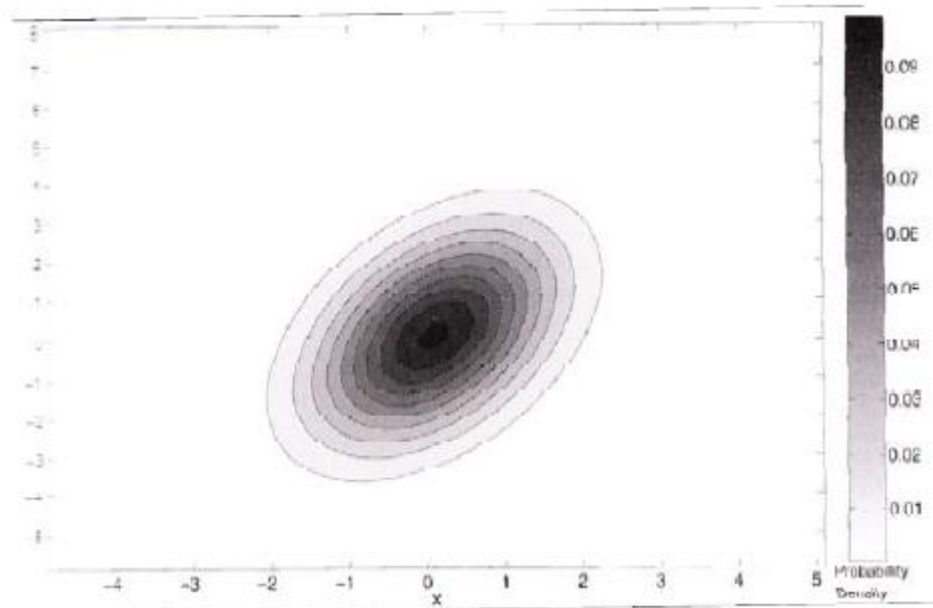
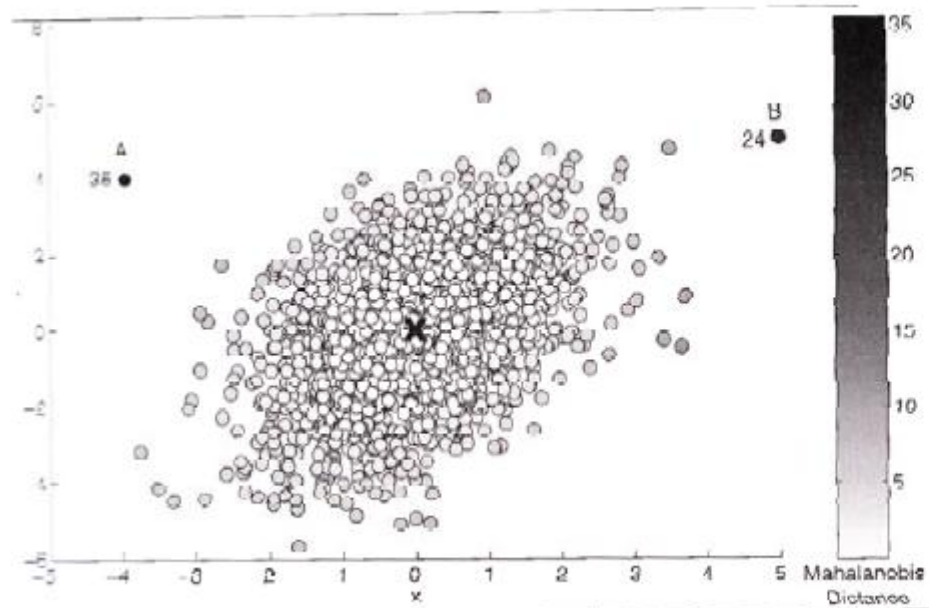
where

- $\mathbf{x}$ – observation vector
- $\bar{\mathbf{x}}$ – **sample mean vector**
- $\mathbf{S}$ – **sample covariance matrix**
- $\mathbf{S}^{-1}$ – inverse of the sample covariance matrix

Unlike Euclidean distance, Mahalanobis distance:

- accounts for **different scales of variables**
- accounts for **correlations between variables**

Multivariate Outliers: Mahalanobis Distance



Interpretation

- small $D^2 \rightarrow$ observation close to the center of the dataset
- large $D^2 \rightarrow$ observation far from the data cloud $\rightarrow$ potential outlier

Multivariate Outliers: Mahalanobis Distance

Key Property

- For a dataset with **p variables**:

$$D^2 \approx \chi^2(p)$$

where

- **D^2** – squared Mahalanobis distance
- **p** – number of variables

Decision Rule

- An observation is considered an outlier if:

$$D^2 > \chi^2(\alpha, p)$$

where

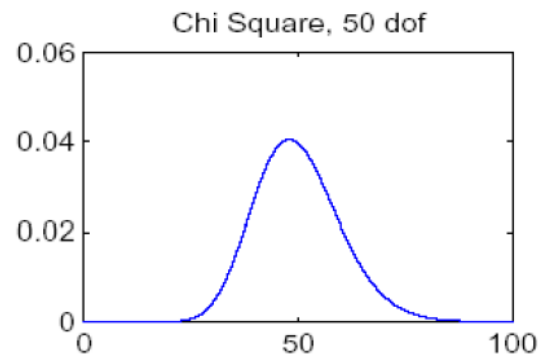
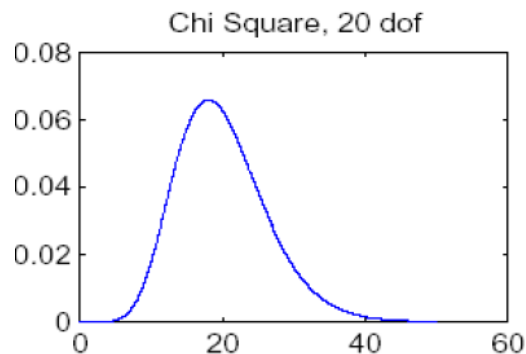
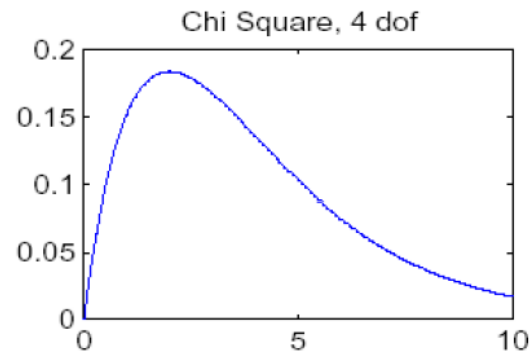
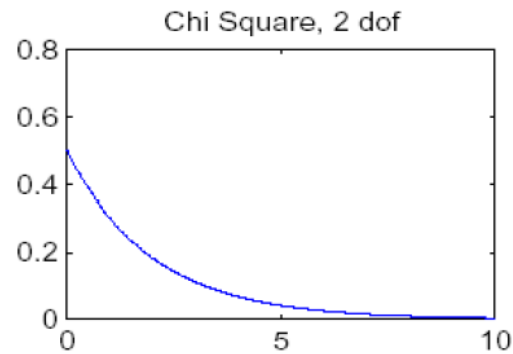
- **α** – significance level (e.g., 0.05)
- **p** – number of variables

Multivariate Outliers: Mahalanobis Distance

Problems

- Curse of dimensionality

The larger the degree of freedom, the more similar the *MDist* values for all points



x-axis: observed D^2 values
y-axis: frequency of observation

Multivariate Outliers: Mahalanobis Distance

Robustness

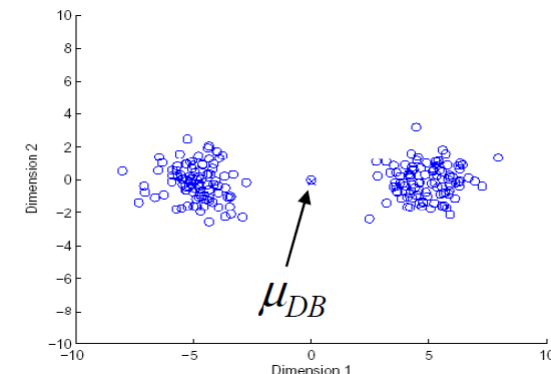
- Means and (co)variances are very sensitive to outliers
- These values are computed for the complete data set (including potential outliers)
- The *Mahalanobis distance* is used to determine outliers although its values are influenced by these outliers

=> Minimum Covariance Determinant [Rousseeuw and Leroy 1987]

minimizes the influence of outliers on the Mahalanobis distance (uses a subset of points which minimizes the determinant of variance-covariance matrix. Compute mean and S based on this subset)

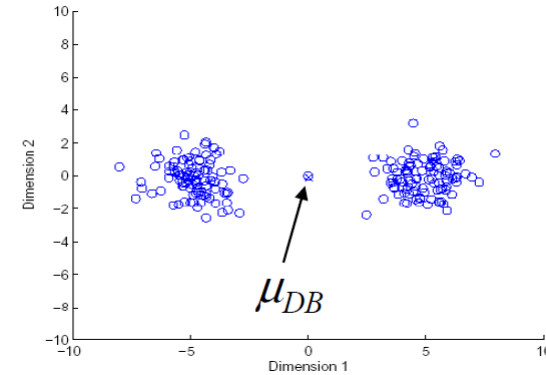
Discussion

- Data distribution is fixed
- Global method
- Outputs a label but can also output a score
- Low flexibility (no mixture model)



Multivariate Outliers: Using Mixture of Parametric Distributions

- Assuming data generated by a normal distribution could be sometimes overly simplified
- Example (right figure): The objects between the two clusters cannot be captured as outliers since they are close to the estimated mean



Assume the data consist of **k clusters**, each approximately following a multivariate normal distribution

1. Identify clusters using k-means or Gaussian mixture model.
2. For each cluster j estimate:
 - mean vector: μ_j
 - covariance matrix: Σ_j
3. For an observation $\mathbf{x}$, compute:
$$D_j^2(\mathbf{x}) = (\mathbf{x} - \mu_j)^T \Sigma_j^{-1} (\mathbf{x} - \mu_j)$$

This measures how far the observation is from the center of cluster j .

4. Select the smallest distance.
5. Compare with the **χ^2 threshold** to detect outliers.

Multivariate Outliers: Stahel–Donoho Outlyingness

Designed to detect outliers even when:

- data contain extreme values
- classical estimators (mean, covariance) are unreliable
- the data distribution is unknown

The method is based on **robust projections of the data onto many directions**.

Basic Idea

- For each observation $\mathbf{x}$, the algorithm measures how extreme it appears when the data are projected onto many directions.
- The outlyingness of $\mathbf{x}$ is defined as the **largest standardized deviation over all directions**.

Multivariate Outliers: Stahel–Donoho Outlyingness

1. For a direction $\mathbf{a}$, project the data:

$$\mathbf{y} = \mathbf{a}^T \mathbf{x}$$

2. The standardized deviation along this direction is:

$$|\mathbf{a}^T \mathbf{x} - \text{median}(\mathbf{a}^T \mathbf{X})| / \text{MAD}(\mathbf{a}^T \mathbf{X})$$

3. The **Stahel–Donoho outlyingness** of $\mathbf{x}$ is:

$$\text{SDO}(\mathbf{x}) = \max_{\mathbf{a}} |\mathbf{a}^T \mathbf{x} - \text{median}(\mathbf{a}^T \mathbf{X})| / \text{MAD}(\mathbf{a}^T \mathbf{X})$$

Interpretation

- small **SDO**($\mathbf{x}$) $\rightarrow$ observation consistent with the bulk of the data
- large **SDO**($\mathbf{x}$) $\rightarrow$ potential multivariate outlier

Multivariate Outliers: Stahel–Donoho Outlyingness

In practice, computing the maximum over **all possible directions** is not feasible. Therefore the algorithm uses a **large set of random directions**.

Algorithm

1. Generate many directions $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$.
Often directions are defined by pairs of observations.
2. For each direction $\mathbf{a}$:
 - project the data: $\mathbf{z}_i = \mathbf{a}^T \mathbf{x}_i$
 - compute robust location and scale:
 - $\text{median}(\mathbf{z})$
 - $\text{MAD}(\mathbf{z})$
3. For each observation $\mathbf{x}_i$, compute
 - $\text{SDO}(\mathbf{x}_i) = \max_{\mathbf{a}} |\mathbf{a}^T \mathbf{x}_i - \text{median}(\mathbf{a}^T \mathbf{X})| / \text{MAD}(\mathbf{a}^T \mathbf{X})$
4. Observations with **large SDO values** are considered outliers.

Multivariate Outliers: Stahel–Donoho Outlyingness

How many outliers?

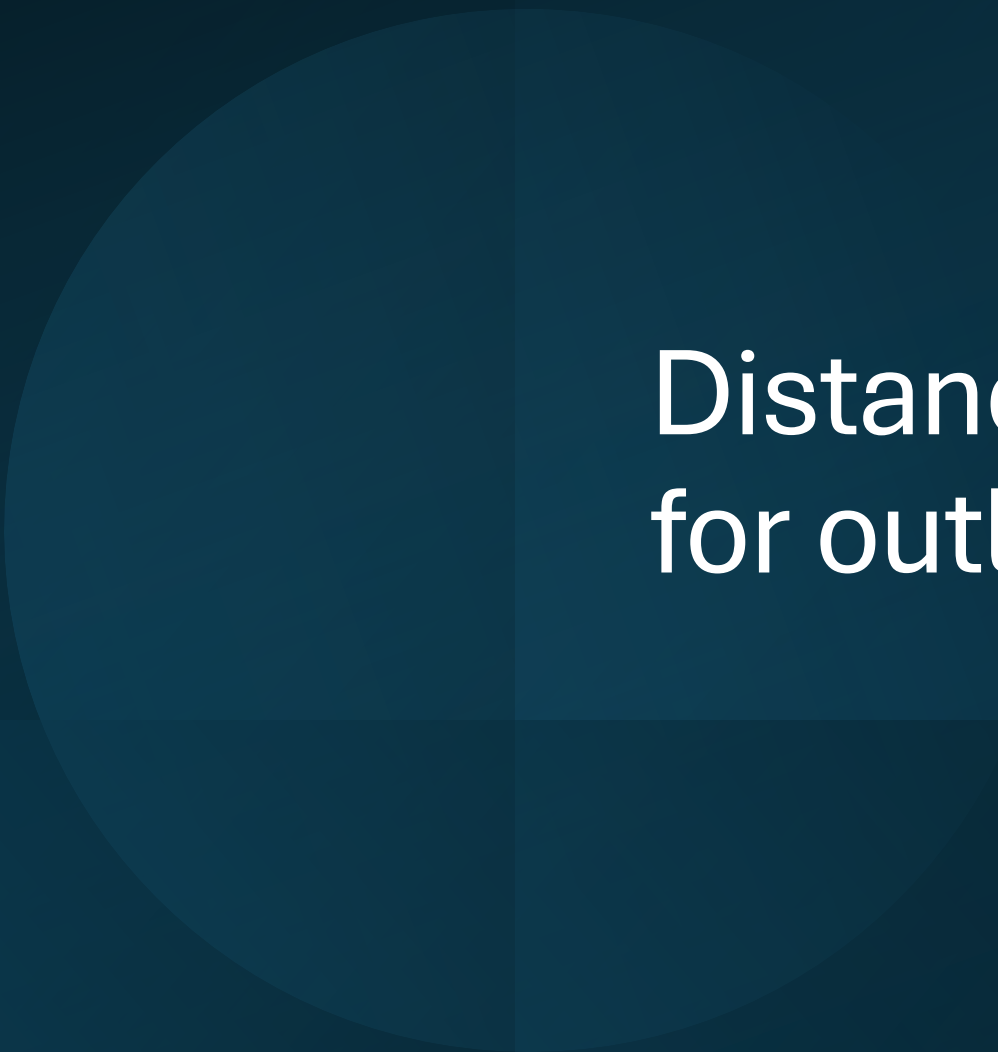
Process the n outlyingness scores

- Choose the samples that exceed $\text{median} + 3 \cdot \text{MAD}$
- Choose top 1–5% or top k
- Visualize ordered outlyingness values on a line plot and look for an elbow

Multivariate Outliers: Stahel–Donoho Outlyingness

Key Characteristics

- uses **robust statistics** (median, MAD)
- does not assume **multivariate normality**
- works well when classical Mahalanobis distance fails
- effective in **high-dimensional data**



Distance-based methods for outlier detection

Distance-Based Outlier Detection

Detect outliers based on the **relative distance between data points**, without making assumptions about the underlying distribution.

Basic Idea

- An observation is considered an outlier if it is far away from most other observations in the dataset.
- > Objects that are isolated in the data space are potential outliers.

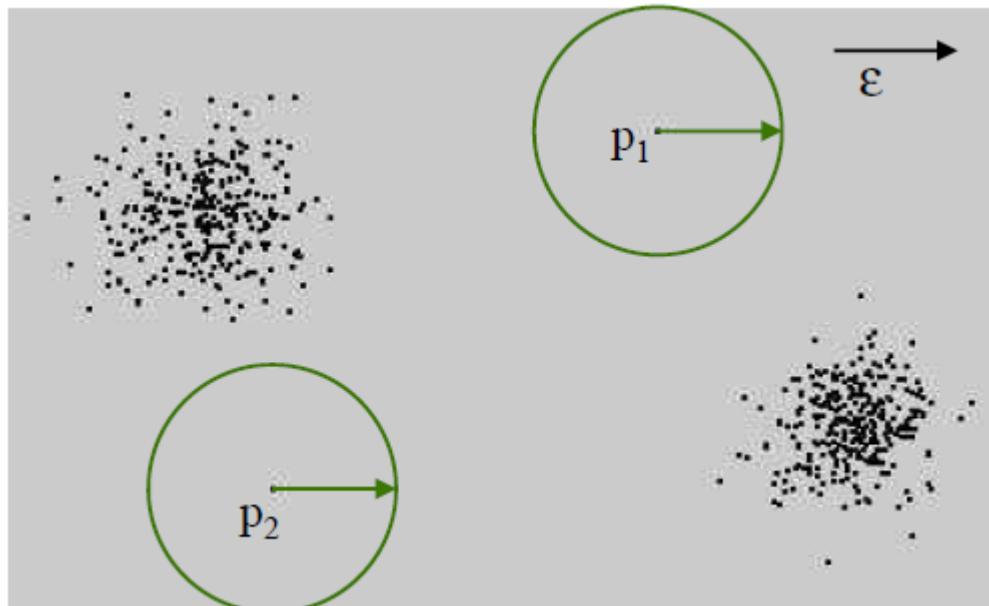
Distance-Based Outlier Detection

(ϵ, π) -model (Knorr and Ng, 1997)

Formal Definition

- A point **p** in a dataset of **N observations** is a **distance-based outlier** if at most $\pi \cdot N$ of all other points have a distance to p less than ϵ

$$OutlierSet(\epsilon, \pi) = \{p \mid \text{Card}(\{q \mid \text{dist}(p, q) < \epsilon\}) \leq \pi \cdot N\}$$



Interpretation

For each point **p**:

1. Consider a **neighborhood of radius ϵ** around **p**.
 2. Count the number of points within this neighborhood.
- If the number of neighbors is **too small**, the point **p** may be an **outlier**.

Distance-Based Outlier Detection

kNN distances

Idea

- Points that are **far from their nearest neighbors** are more likely to be outliers.

[Ramaswamy et al, 2000]: distance from p to its k -th nearest neighbor

$$\text{score}(p) = \text{dist}(p, \text{NN}_k)$$

[Angiulli and Pizzuti 2002]: average distance to the k nearest neighbors

$$\text{score}(p) = (1 / k) \cdot \sum \text{dist}(p, \text{NN}_j)$$

Distance-Based Outlier Detection

kNN graph (Hautamaki et al. 2004)

Basic Idea

- Construct the k NN graph for a data set
 - Vertices: data points
 - Edge: if $q \in k\text{NN}(p)$ then there is a directed edge from p to q
- A vertex that has an indegree less than equal to T (user defined threshold) is an outlier

The indegree of a vertex in the k NN graph equals to the number of reverse kNNs (RkNN) of the corresponding point

- The RkNNs of a point p are those data objects having p among their k NNs
- Intuition of the model: outliers are
 - points that are among the k NNs of less than T other points
 - have less than T RkNNs

Distance-Based Outlier Detection

Discussion

- Output can be a scoring (kNN-distance models) or a labeling (kNN graph model and the (ϵ, π) -outlier model)
- Approaches are local (resolution can be adjusted by the user via ϵ or k)

Advantages

- simple and intuitive
- no assumption about data distribution
- applicable to many types of data

Limitations

- computationally expensive for large datasets
- sensitive to the choice of parameters
- may fail when **local densities vary strongly**



Density-Based methods for Outlier Detection

Density-Based Outlier Detection

- Distance-based methods assume that **outliers are far from most other points**.
- However, this assumption may fail when the dataset contains **regions with different densities**.
- In such cases, a point may be far from many points but still belong to a **sparse cluster**.
- Density-based methods address this problem by comparing the **local density of a point** with the density of its neighbors.

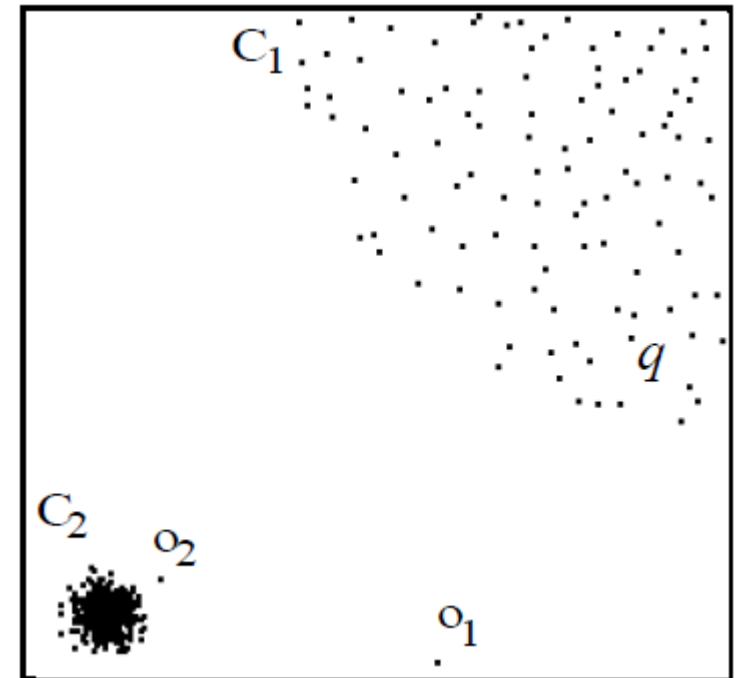
Example

– (ϵ, π) -model

Parameters ϵ and π cannot be chosen so that o_2 is an outlier but none of the points in cluster C_1 (e.g. q) is an outlier

- Outliers based on kNN-distance

kNN-distances of objects in C_1 (e.g. q) are larger than the kNN-distance of o_2



Local Outlier Factor (LOF)

[Breunig et al. 1999], [Breunig et al. 2000]

Basic Idea

- A point is considered an **outlier** if its **local density is significantly lower** than the density of surrounding points.
- > Points located in **low-density regions surrounded by higher-density regions** are potential outliers.

Local Outlier Factor (LOF) Definitions

Let p, o be points in the dataset.

- **k-distance(p)**

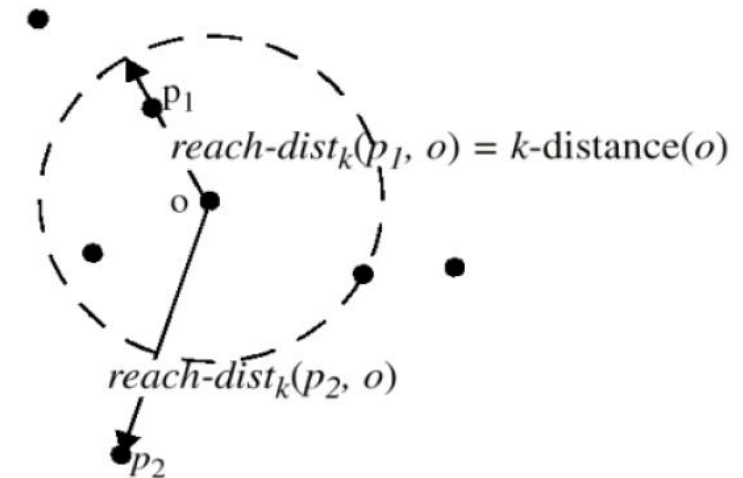
The k-distance of p is the distance between p and its k-th nearest neighbor.

- **Reachability Distance** - reduces the influence of very small distances between points.

$$\text{reach-dist}_k(p, o) = \max \{ k\text{-distance}(o), \text{dist}(p, o) \}$$

Interpretation

- Reachability of p from o - asymmetric
- If p is very close to o , the distance is replaced by **k-distance(o)**
- This prevents extremely small distances from dominating the density estimation
- Reachability distance ensures that density estimates are **more stable**.



Local Outlier Factor (LOF) Definitions

- **k-neighborhood of p - $N_k(p)$**

$$N_k(p) = \{ o \mid \text{dist}(p, o) \leq k\text{-distance}(p) \}$$

$N_k(p)$ contains at least k neighbors, but possibly more if multiple points lie at the same distance as the k -th nearest neighbor.

- **Local Reachability Density (LRD)** - estimates the density around point p .

$$\text{LRD}_k(p) = 1 / ((1 / |N_k(p)|) \cdot \sum \text{reach-dist}_k(p, o))$$

where the summation is taken over all $o \in N_k(p)$.

Interpretation

- small reachability distances → **high density**
- large reachability distances → **low density**

Local Outlier Factor (LOF) Definitions

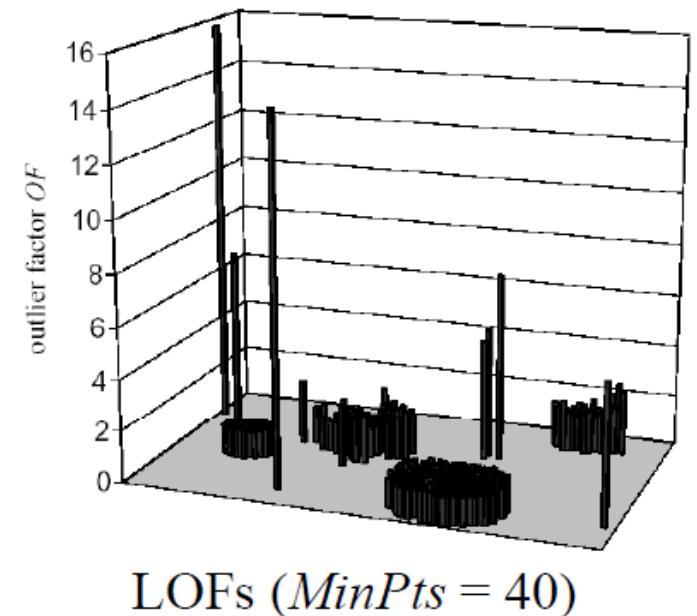
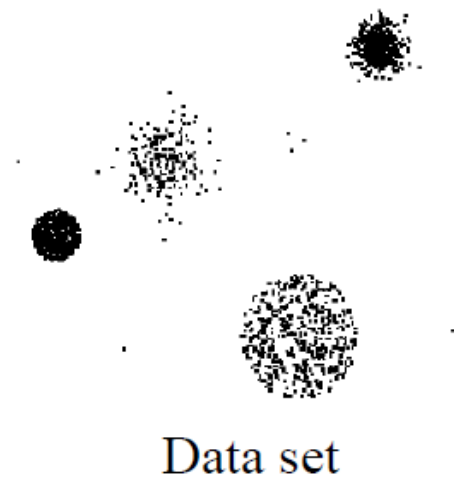
- **Local outlier factor (LOF)**

$$\text{LOF}_k(p) = (1 / |N_k(p)|) \cdot \sum (\text{LRD}_k(o) / \text{LRD}_k(p))$$

where the summation is taken over all $o \in N_k(p)$

Outliers are points whose local density is significantly **lower than the density of their neighbors**:

- $\text{LOF} \approx 1$ -> point has density similar to neighbors
- $\text{LOF} > 1$ -> point is less dense than neighbors
- $\text{LOF} \gg 1$ -> strong outlier



Local Outlier Factor (LOF)

Pros and Cons

Advantages

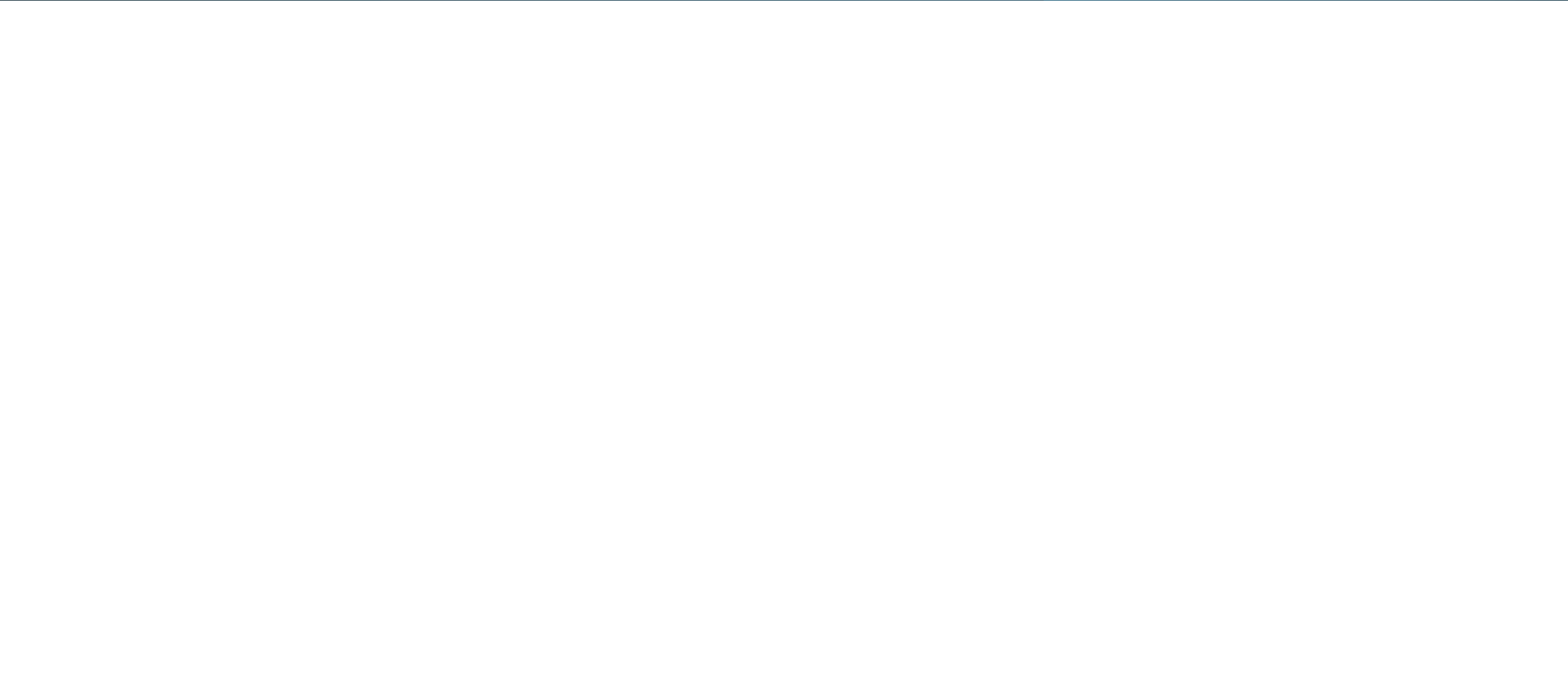
- detects **local outliers**
- works well with **datasets containing clusters of different densities**
- does not assume a specific data distribution

Limitations

- requires selecting parameter **k**
- computationally expensive for large datasets
- performance may degrade in **high-dimensional data**



Machine learning methods for Outlier Detection



Bibliography